Hugo Marquerie 13/01/2026

Prop transformada fourier traslacion modulacion dilatacion

Proposición 1. *Sea* $f \in \mathcal{L}^1(\mathbb{R})$, $y, z \in \mathbb{R}, \delta > 0$, *entonces*

(1) $(T_y f)^\wedge(\xi) = e^{-2\pi i y\xi} f^\wedge(\xi) = M_{-y} f^\wedge(\xi)$;

(2) $(M_z f)^\wedge(\xi) = f^\wedge(\xi - z) = T_z(f^\wedge)(\xi)$;

(3) $(D_\delta f)^\wedge(\xi) = \frac{1}{\delta} f^\wedge\left(\frac{\xi}{\delta}\right) = \frac{1}{\delta} D_{\frac{1}{\delta}}(f^\wedge)(\xi)$;

donde $f^\wedge$ *denota la transformada de Fourier de* f.

***Demostración*:**

(1) Calculamos desde las definiciones y hacemos el cambio de variable $u = x - y$:

$$\begin{aligned}(T_y f)^\wedge(\xi) &= \int_{\mathbb{R}} f(x-y) e^{-2\pi i x\xi}\,\mathrm{d}x = \int_{\mathbb{R}} f(u) e^{-2\pi i (u+y)\xi}\,\mathrm{d}u \\ &= e^{-2\pi i y\xi} \int_{\mathbb{R}} f(u) e^{-2\pi i u\xi}\,\mathrm{d}u = e^{-2\pi i y\xi} f^\wedge(\xi) = M_{-y} f^\wedge(\xi).\end{aligned}$$

(2) En este caso, no es necesario un cambio de variable:

$$(M_z f)^\wedge(\xi) = \int_{\mathbb{R}} e^{2\pi i z x} f(x)\, e^{-2\pi i x\xi}\,\mathrm{d}x = \int_{\mathbb{R}} f(x)\, e^{-2\pi i x(\xi - z)}\,\mathrm{d}x = f^\wedge(\xi - z) = T_z(f^\wedge)(\xi).$$

(3) Hacemos el cambio de variable $u = \delta x$:

$$\begin{aligned}(D_\delta f)^\wedge(\xi) &= \int_{\mathbb{R}} f(\delta x)\, e^{-2\pi i x\xi}\,\mathrm{d}x = \int_{\mathbb{R}} f(u)\, e^{-2\pi i \frac{u}{\delta}\xi}\, \frac{1}{\delta}\,\mathrm{d}u \\ &= \frac{1}{\delta}\int_{\mathbb{R}} f(u)\, e^{-2\pi i u \frac{\xi}{\delta}}\,\mathrm{d}u = \frac{1}{\delta} f^\wedge\left(\frac{\xi}{\delta}\right) = \frac{1}{\delta} D_{\frac{1}{\delta}}(f^\wedge)(\xi).\end{aligned}$$

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Referenciado en

- `teo-inversion-transformada-fourier`

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